

Spatial Cluster Randomized Trials - ~~Efficient estimation~~ with spillover effects & spatial dependence

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Abstract

Background: Investigators designing cluster randomized trials desire insights directing treatment assignment methodology for studies in the presence of spillover effects and spatial dependence.

Methods: Treatment assignment methods including random sampling and block stratified are applied and spatial autoregressive modeling is applied with consideration for spillover effects and spatial dependence for estimation of intervention effects.

Simulation: Simple Random sampling & Block Stratified Sampling treatment assignment methods were applied to spatial grids with dimensions of 2x4, 3x3, and 3x4 clusters (cells). Varieties of spillover effects and spatial dependence sampled observations were considered for estimation of the intervention effect via a spatial autoregressive (SAR) model.

Results: Findings of an extensive simulation study considering the comparison of block stratified and random treatment assignment sampling methods with consideration of spillover effects, spatial dependence, and regional spillover restrictions indicate that randomly selected treatment assignments result in reduced average error when estimating intervention effects, but block stratified treatment assignments lead to reduced variation among a series of estimates. This is an indication a block stratified treatment arrangement won't achieve the minimal level of estimation error, but it remains consistent across a range of selected parameters.

Conclusion: A clear relationship between spillover effect magnitude and MSE for intervention effect estimation is apparent. MSE for the intervention effect is minimized for random sampling treatment assignment, but block stratified assignment minimizes error variance.

1 Introduction

1.1 Background and Relevance

- **NEED TO DO LITERATURE REVIEW**
- introduce randomized trials and cluster randomized trials [CITE]
- emphasize the utility of randomized trials and need for direction with design decisions [CITE]
- acknowledge challenges with spillover and spatial dependence [CITE]

1.2 Motivation

- discuss the gap in existing research and the needs that investigators have when designing Spatial CRTs [CITE]
- there is a gap/need for simple/intuitive recommendations on how to construct sampling of treatments when designing a Spatial CRT
- how do existing studies address sampling? what do existing spatial CRT papers talk about? Point out that there is a "hole" and nobody has talked about sampling methods. Has anybody consider BOTH spillover and spatial dependence concurrently? [CITE]

1.3 Aims & Contributions

This study aims to investigate the efficiency of treatment assignment sampling methodologies in spatial cluster randomized trials for estimating the intervention effect in the presence of both spatial spillover effects with consideration for spatial dependence. We compare two different sampling approaches for treatment assignment on clusters: Simple Random Sampling (SRS) and Block Stratified Sampling (BSS). BSS specifically seeks to isolate clusters assigned as intervention or control from other clusters of the same treatment assignment. Our simulation study will explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence and absence of spatial dependence, and varying restrictions on clusters where spillover is permitted to occur. A strong understanding of patterns that exist across these sampling methods and other features considered provides practical guidance to investigators designing Spatial CRTs, assisting them with selecting sampling strategies that align with their goals to minimize bias and/or variance of intervention effect estimates in a spatial setting.

1.4 Overview

The remainder of this article is organized as follows: Section 2 outlines the theoretical framework of the study, including the data generation process, sampling procedures, and estimation methods employed. Section 3 details algorithm and results of the simulation study undertaken to evaluate the sampling methods considered. Section 4 considers a real-world application of these findings to make a recommendation for other investigators. Finally, Section 5 discusses the implications of our findings, potential drawbacks to the work performed, and potential areas for continued future research.

2 Methods

2.1 Cluster Randomized Trials

AN INTRODUCTION TO RANDOMIZED TRIALS AND CLUSTER RANDOMIZED TRIALS HERE [CITE]

2.2 Neighbor Definitions

2.2.1 Contiguity-Based

Contiguity-based neighbors[7] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other clusters (or cells) that share a common boundary. This boundary may be an edge and/or vertex. The **spdep** R package[4] includes functions for dealing with spatial neighbors and spatial dependence structures.

Rook: Rook neighbors fulfill a contiguity condition where a pair of cells share a common boundary made up by an edge.

Queen Queen neighbors fulfill a contiguity condition where a pair of cells share a common boundary of either an edge or a vertex. Neighbors may also share both an edge and a vertex.

Bishop Neighbors that share a common boundary at the vertex (corners) of the spatial region are known as Bishop neighbors[7].

Higher-Order Neighbors Spatial regions that do not directly share an edge and/or vertex with another region, but instead share an edge and/or vertex with neighbors of a region are higher-order neighbors. Figure 4 illustrates a comparison between the first order neighbors for rook and queen contiguity (top row) versus the second order neighbors for rook and queen contiguity. These higher-order neighbors are “neighbors of neighbors” of the reference cell.

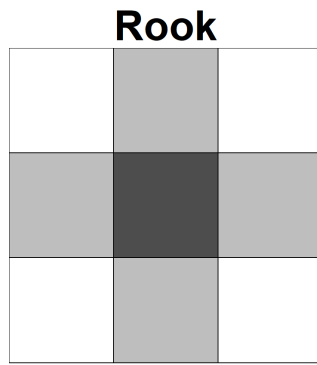


Figure 1: Rook neighbors (light) share an edge with a reference spatial region (dark)[12]

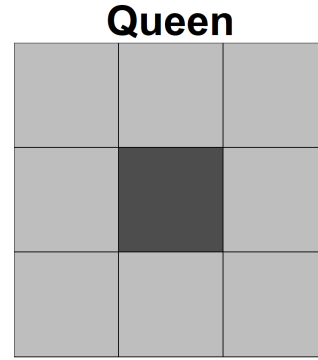


Figure 2: Queen neighbors (light) share either an edge or vertex (or both) with a reference spatial region (dark)[12]

Figure 3: Comparison of Rook and Queen Contiguity Definitions

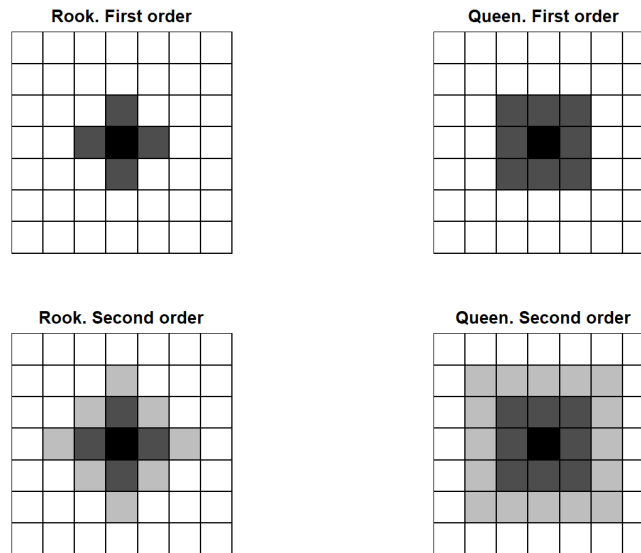


Figure 4: Higher order neighbors (light) are “neighbors of neighbors” of a reference cell (darkest)[12]. Rook neighbors (left) share edge boundaries and Queen neighbors (right) share either edge or vertex boundaries.

2.2.2 Distance-Based

Distance-based neighbors[7] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other areas (or cells) that lie within a specified distance of another cell or observation. This distance is commonly delineated by the radius of a circle around the center of a reference spatial region. The region specified in Figure 5 identifies distance-based neighbors of the reference cell (dark) as neighboring cells where the centroid of the cells lie within the circumscribed boundary around the reference cell. In addition to cells, distance-based neighborhoods can also be constructed based on individual points that are within a specified distance of another point.

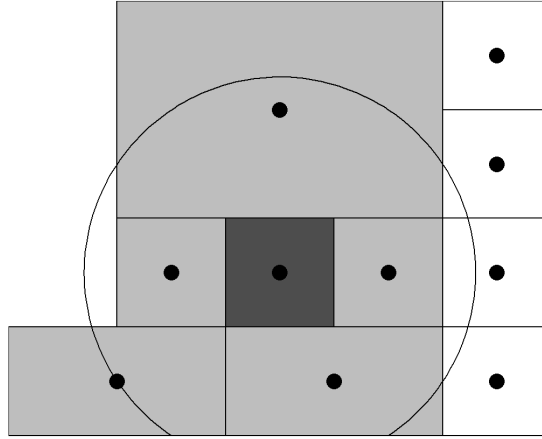


Figure 5: Distance-based neighbors (light) have centroids within a fixed radius of the reference cluster (dark). Clusters with centroids beyond this distance are not considered as neighbors (white)[12]

2.3 Intervention & Spillover Effects

Two primary effect types are considered to understand how treatments affect subjects where they are applied and other subjects in the surrounding areas. The intervention effect and spillover effect are two parameters that are estimated in this study. Spatial cluster randomized trials have subjects or regions that are assigned to receive an intervention and others who are assigned as controls that do not receive an intervention. Observation of the downstream outcomes allow for estimation of the direct intervention effect and indirect spillover effect on the study region.

2.3.1 Intervention Effect

The intervention effect (β) represents the estimated effect of the direct intervention applied in a study. This is a measure of the study outcome that is directly attributed to the intervention that is assigned and applied in the study. Estimation of this intervention effect is possible due to the presence of subjects or regions assigned as a control (that do not receive an intervention) for comparison. Over an array of applications the intervention effect could be a vaccine to prevent infectious disease, a training program for government employees, an educational program for students, and many more.

2.3.2 Spillover Effect Methods

The spillover effect (ψ) represents an indirect effect observed from the intervention on subjects or regions where an intervention was not directly applied[2][3]. There are a number of ways to model a spillover effect as it is observed due to spatial proximity to subjects or regions where an intervention is applied. In this study, only binary spillover effects were considered, but some alternative spillover methods are also discussed. In the following definitions, x_i is an indicator of whether subject i receives the intervention, z_i is an indicator of whether the same subject neighbors another subject that receives the intervention (1) or not (0) and is eligible to experience spillover, $d_{\max}$ is the diameter of each cluster of uniform size, d_i is the distance that a subject is from the nearest intervention source, and ψ is the maximum proportion of the intervention effect that can be experienced as spillover.

- **Binary:** a fixed level of a spillover effect is applied if a subject or region is deemed to be a neighbor of a subject or region where the intervention is applied. This response for subject i considering this spillover effect is expressed as

$$y_i = \alpha + \beta x_i + \psi z_i + \varepsilon_i.$$

- **Linear:** The effect of an intervention can decay as distance from the treatment sources increases. A linear decay spillover effect can be applied such that subjects closer to others receiving an intervention experience a greater spillover effect (larger proportion of the intervention effect) and those a greater distance from an intervention source receive a lesser effect (or none) from the intervention. The response for a subject i with a linear decay spillover effect is

$$y_i = \alpha + \beta x_i + \psi \left(1 - \frac{d_i}{d_{\max}} \right) z_i$$

- **Categorical:** A categorical decay spillover effect models a step-wise drop-off in the proportion of the intervention effect experienced at increasing levels of distance from the nearest intervention source. Contrary to the linear spillover effect, a categorical spillover effect allows for a constant effect to be experienced at concentric bands of distance from an intervention source. Subject i has the following response with consideration of a categorical spillover effect

$$y_i = \alpha + \beta x_i + \psi \left[I \left(d_i \in \left[0, \frac{1}{4} d_{\max} \right] \right) + 0.75 I \left(d_i \in \left[\frac{1}{4} d_{\max}, \frac{1}{2} d_{\max} \right] \right) \right. \\ \left. + 0.50 I \left(d_i \in \left[\frac{1}{2} d_{\max}, \frac{3}{4} d_{\max} \right] \right) + 0.25 I \left(d_i \in \left[\frac{3}{4} d_{\max}, d_{\max} \right] \right) \right] z_i,$$

Here, each of the domains of d_i denotes a “concentric” ring of distance from the intervention zone that receives a certain proportion of the intervention effect (decays in step with distance from the intervention zone). The rings here are the first, second, third, and fourth quartiles of distance from an intervention boundary with respect to the diameter of a cluster.

- **Exponential:** An exponential decay pattern of spillover has a much “steeper” drop-off in the effect experienced by a subject neighboring an intervention cluster as the distance from the intervention source increases. This is a reasonable model for spillover if subjects nearby an intervention are expected to experience a strong spillover effect, but subjects that a marginally larger distance and beyond experience little to no effect in the form of spillover. The response for subject i with an exponential decay spillover effect is

$$y_i = \alpha + \beta x_i + \psi \left[\exp \left(-k \frac{d_i}{d_{\max}} \right) \right] z_i,$$

where k denotes the decay rate of the exponential function.

- **Gaussian:** A Gaussian decay spillover appears as a “bell”-shaped distribution when the effect strength is plotted against distance. This decay pattern is similar to exponential decay, but the effect decay is much slower at first before a steep drop-off in effect strength occurs at a moderate to large distance from the intervention source. The response for subject i with a gaussian decay spillover effect (and spread parameter σ) is

$$y_i = \alpha + \beta x_i + \psi \exp \left(-\frac{d_i^2}{2\sigma^2} \right) z_i,$$

where small σ correspond to a rapid decay of the spillover effect and large σ allow for a much larger distance of spillover effect decay.

2.3.3 Spillover Effect Application

In this study, there are two specific ways in which spillover of the applied intervention is allowed to propagate to other cells or subjects. In both cases, spillover effects originate from areas where the original intervention treatment is applied. The first case allows for spillover to propagate from cells assigned as receiving the intervention into only cells assigned as control cells. In this case, the effect of

the intervention is not permitted to spillover into other intervention cells. This suggests a case where the intervention effect applied is the maximum observable effect and any potential spillover cannot increase the observed effect of a particular intervention. Another method of spillover application is considered where spillover effects are permitted to be applied to both control AND intervention cells that are neighbors of a cell that receives an intervention. This application suggests that an effect greater than that of the original intervention applied can be potentially be observed in a cell that received an intervention and also had spillover from another intervention cell. It is also to be expected that any change in the estimated effects would not be observed in spatial settings where the treatment allocation does not satisfy block stratification conditions (where no intervention cells are rook neighbors with other intervention cells). A representative comparison of these two methods of permissibility of spillover effects is given in Figure 6.

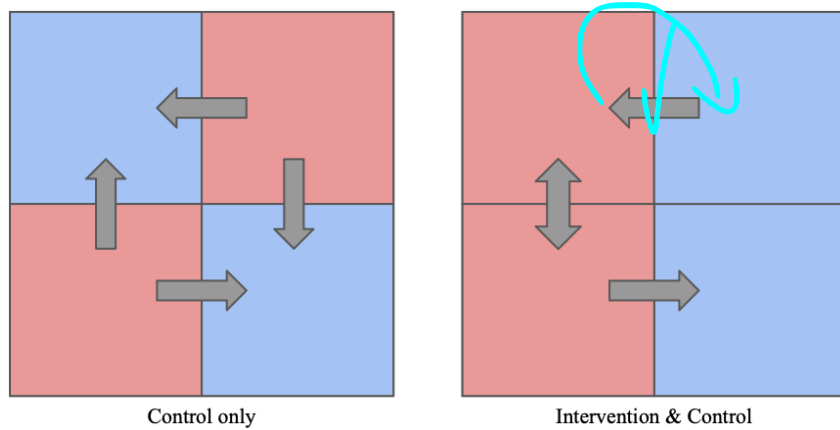


Figure 6: (Left) Spillover application from intervention cells (red) to only control cells (blue) & (Right) spillover application to intervention (red) and control cells (blue).

2.4 Sampling Methods - Random & Block Stratification

In this study, two different treatments are considered: intervention & control. Assignment of these treatments is considered under two methods: Simple Random Sampling (SRS) & Block Stratified Sampling (BSS). Random sampling treatment assignment considers all possible combinations of intervention & control arrangements on a specified spatial grid and provides equal probability of assignment for any given combination of intervention and control. For example, a 2 row by 3 column spatial grid has 6 total cells. Consideration of a 1:1 ratio between intervention and control treatments provides that three cells are assigned as intervention and three cells are assigned as control. Therefore, there are $\binom{6}{3} = 20$ different combinations of treatments for this specific spatial region.

Block stratified sampling seeks to ensure that for a given balanced number of intervention and control cells, that no intervention or control cells share a common edge boundary with each other. An exception to this rule occurs in spatial setups where both dimensions are odd (ie. 3x3) such that the count of intervention and control cells must be out of balance and we assert that the control class will always be in the majority. In these instances, we relax the common edge boundary restriction for control cells and only require that intervention cells not share a common edge boundary with each other in order to satisfy the block stratification condition.

We now consider a series of spatial grid setups to address efficient estimation of the intervention effect. These spatial grids are denoted as (row $\times$ column) in terms of dimension and we seek to apply a balanced distribution between intervention and control cells where possible. When this isn't possible, we defer to the control group being the majority class (more cells) as mentioned previously.

2.4.1 2x4 Grid

One representative spatial setting considered is grid of 8 individual cells laid out in an arrangement of 2 rows and 4 columns. For this grid setup, a balanced set of intervention and control cells can be assigned where 4 cells are assigned as intervention regions and 4 cells are assigned as control regions. For a grid of 8 total cells and 4 cells of each treatment type, there are $\binom{8}{4} = 70$ unique arrangements that cells can be assigned between treatment and control. Figure 7 displays one of these combinations that satisfies the block stratified condition where intervention and control cells do not have any rook neighbors of the same treatment type. For this spatial setting, there are 2 (out of 70) combinations that satisfy the block stratification condition. Figure 8 displays an example of the 2x4 grid spatial setting with a randomly sampled arrangement of intervention and control cells. Note that both treatment types have cells that are rook neighbors with the same treatment.

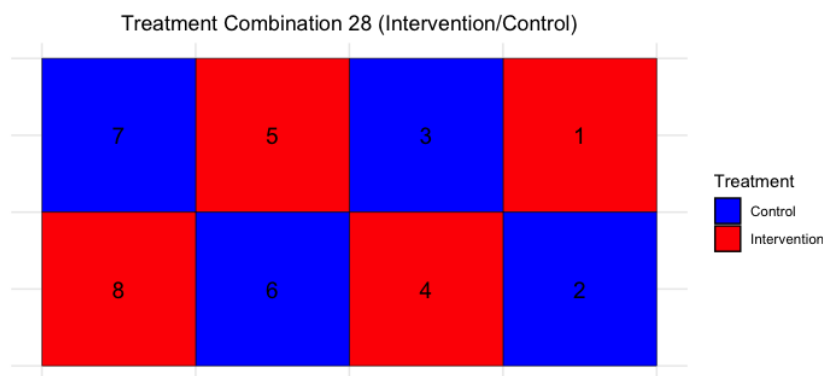


Figure 7: 2x4 spatial grid of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling

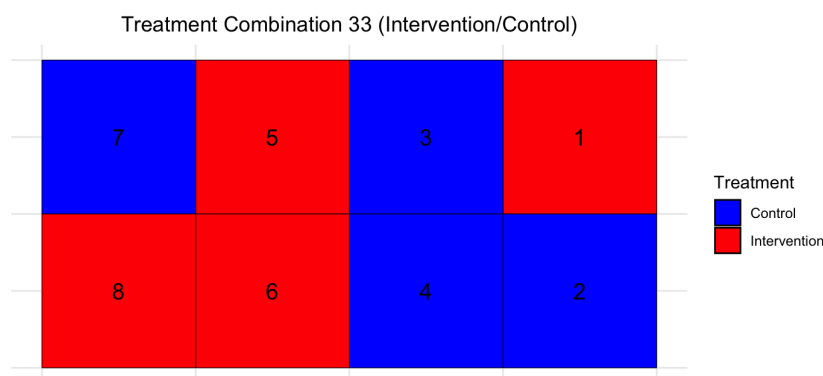


Figure 8: 2x4 spatial grid of intervention (red) and control (blue) clusters assigned via Simple Random Sampling

2.4.2 3x3 Grid

The 3x3 symmetric grid setting of 9 cells allows for $\binom{9}{4} = \binom{9}{5} = 126$ unique combinations of intervention and control cells where 5 cells are allocated as control and 4 cells are assigned as treatment. The majority class of 5 cells is deemed to be the control for the sake of hypothetical resource management. Figure 9 illustrates one combination of treatment and control cells that completely satisfies the block stratification conditions. Additionally, Figure 10 displays another combination of cells that satisfies a relaxed condition of block stratification where control cells are allowed to have other control cells as rook neighbors. This is possible in spatial grid settings where both dimensions (row, column) are odd in length. There is one combination for the 3x3 grid setting that satisfies the strongest block stratification condition and 5 more combinations that satisfy the relaxed block stratification condition for a total of 6. An example of randomly sampled treatment assignment that doesn't satisfy block stratification conditions is provided in Figure 11.

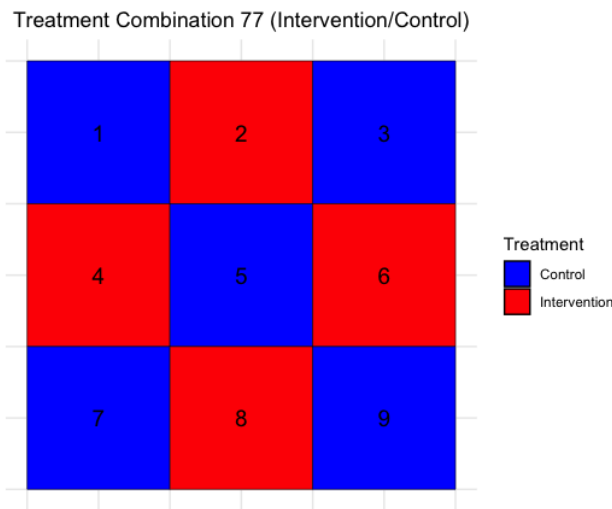


Figure 9: 3x3 spatial grid of clusters with intervention (red) and control (blue) assigned via Block Stratified Sampling

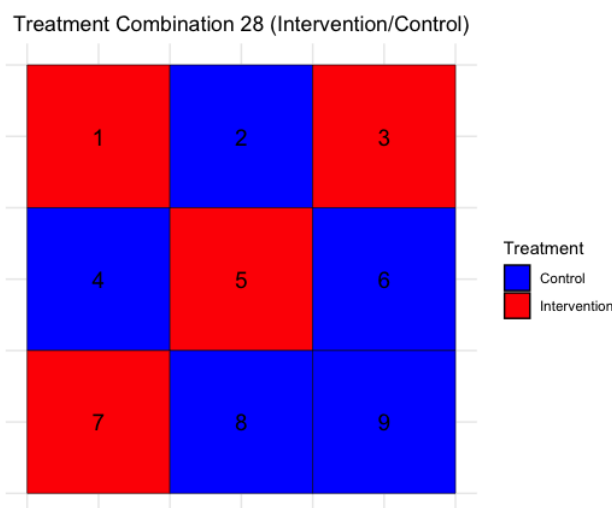


Figure 10: 3x3 spatial grid of intervention (red) and control (blue) clusters assigned via relaxed Block Stratified Sampling

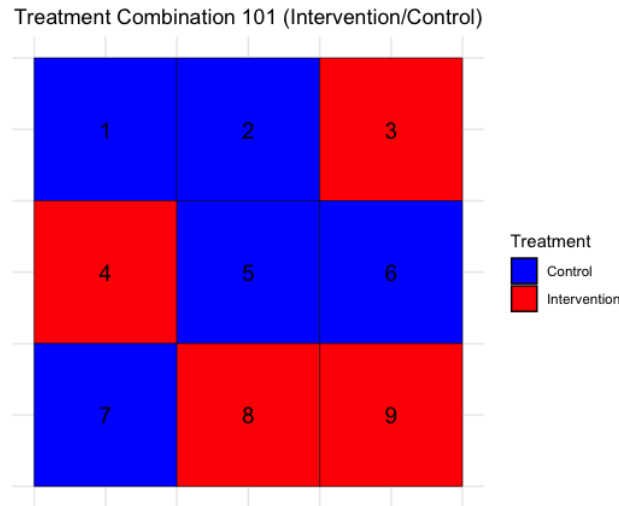


Figure 11: 3x3 spatial grid of intervention (red) and control (blue) clusters assigned via Simple Random Sampling

2.4.3 3x4 Grid

The final spatial setting considered was a 3 row by 4 column (3x4) grid of 12 total cells. This grid, with a balanced allocation of 6 intervention cells and 6 control cells allows for $\binom{12}{6} = 924$ unique arrangements of the two treatments in the spatial region. For this particular grid size, 2 of the 924 treatment combinations satisfy the block stratification condition and Figure 12 presents one of these, with a “checkerboard” design. The remaining treatment combinations are achieved via random sampling of a balanced number of intervention and control cells. Figure 13 illustrates another randomly sampled arrangement of the treatments onto the spatial area where control and intervention cells each have rook neighbors of the same treatment type.

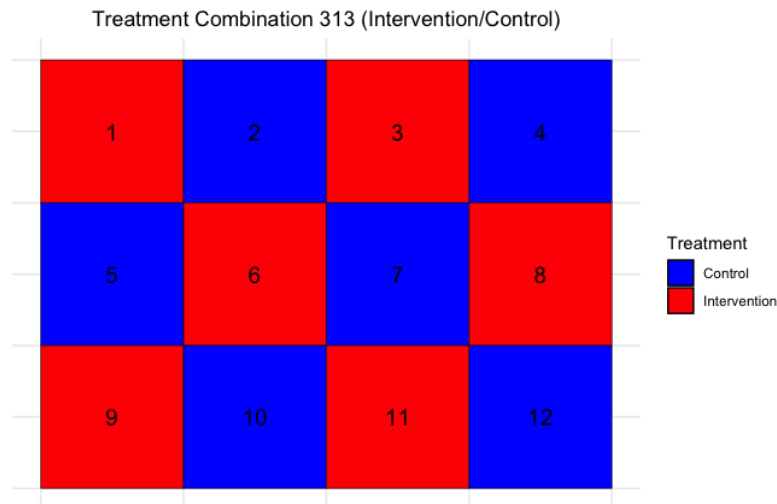


Figure 12: 3x4 spatial grid of clusters with intervention (red) and control (blue) assigned via Block Stratified Sampling

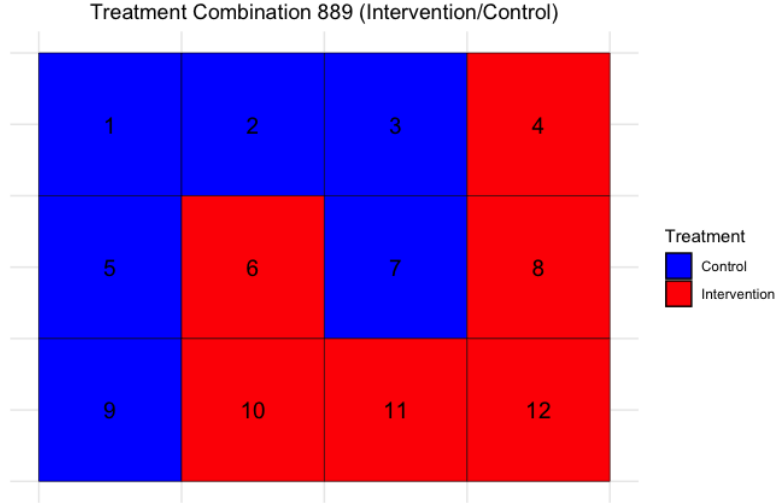


Figure 13: 3x4 spatial grid of intervention (red) and control (blue) clusters assigned via Simple Random Sampling

2.5 Estimation - Spatial Autoregressive (SAR) Model

Spatial dependence of observations can be considered via modeling with a spatial autoregressive model[18][7][17]. A spatial autoregressive model has the form:

$$y_i = \rho \sum_j w_{ij} y_j + \mathbf{x}_i \beta + \varepsilon_i$$

where ρ is the spatial correlation (autoregressive) coefficient and ε_i represents an i.i.d. error term. This can also be expressed in matrix notation as:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \varepsilon$$

The general form of the response for this model for the purpose of this study is

$$y_i = \alpha + \beta x_i + \psi(d_i) z_i + \rho \mathbf{W} \mathbf{y} + \epsilon_i$$

where

- y_i is the observed response for subject i
- α is the baseline/background effect
- β is the intervention effect
- x_i is an indicator of whether subject i receives the intervention
- $\psi(d_i)$ is the spillover effect expressed as a function of distance from the nearest intervention source for subject i
- z_i is an indicator of whether subject i can experience any spillover effect
- $\rho \mathbf{W} \mathbf{y}$ captures spatial dependence from neighboring observations and their effects in the response variable.
 - ρ is a fixed spatial correlation parameter
 - $\mathbf{W}$ is a spatial weights matrix indicating the strength of relationship between neighboring observations

– $\mathbf{y}$ is the response vector

- ε_i is a random error term for each subject

Parameter estimates from the spatial lag model are obtained via maximum likelihood estimation. When considering spatial dependence, the responses are

$$y_i = \rho \sum_{j \neq i} w_{ij} y_j + \alpha + \beta x_i + \psi z_i + \epsilon_i$$

where the residuals are

$$\varepsilon_i = y_i - \rho \sum_{j \neq i} w_{ij} y_j - (\alpha + \beta x_i + \psi z_i).$$

Then, assuming that $\varepsilon_i \sim N(0, \sigma^2)$, the likelihood function is

$$L(\alpha, \beta, \psi, \rho, \sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)^n |\mathbf{I} - \rho\mathbf{W}|}} \exp\left(-\frac{1}{2\sigma^2} \boldsymbol{\epsilon}^\top \boldsymbol{\epsilon}\right)$$

where $|\mathbf{I} - \rho\mathbf{W}|$ is the determinant of $(\mathbf{I} - \rho\mathbf{W})$. The log-likelihood is

$$\log L(\alpha, \beta, \psi, \rho, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) + \log |\mathbf{I} - \rho\mathbf{W}| - \frac{1}{2\sigma^2} \sum_{i=1}^n \left(y_i - \rho \sum_{j \neq i} w_{ij} y_j - (\alpha + \beta x_i + \psi z_i) \right)^2$$

which is maximized with respect to $\alpha, \beta, \psi, \rho$, and σ^2 . In applied simulation tasks, estimates can be obtained via the ‘lagsarlm()’ function from the R spdep library[4].

2.6 Spatial Dependence

Spatial dependence, used interchangeably with spatial autocorrelation, refers to the property of spatial data where observations close to each other tend not be independent of each other. Rather, observations that are close to each other tend to be dependent on each other and influence values of surrounding data. Hence, there is an implied systematic correlation between data at neighboring locations. Spatial correlation can be positive, where similar values tend to be clustered close together, or negative, where dissimilar values are close together. Appropriate consideration of spatial dependence is necessary for performing valid statistical inference as well as for accurate estimation of effects that are modeled. A pair of common methods for assessing spatial autocorrelation are Moran’s I [13][14] and Geary’s C [8][18]. Moran’s I standardizes the spatial auto-covariance by the variance of the data and Geary’s C utilizes the sum of the squared differences between pairs of data as a measure of co-variation. These test statistics are reliant on the spatial weights matrix which is discussed in Section 2.6.3.

2.6.1 Tests of Spatial Dependence - Moran’s I

The Moran’s I statistic quantifies the degree of similarity or dissimilarity between neighboring data using standardized spatial covariance. The statistic takes on values from -1 (negative autocorrelation) to 1 (positive autocorrelation) where the expected value under a null hypothesis of no spatial autocorrelation is $E[I] = -(N-1)^{-1}$, where N is the count of spatial units. Moran’s I has the form

$$I = \frac{N \sum_{i=1}^N \sum_{j=1}^N w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^N \sum_{j=1}^N w_{ij} \right) \sum_{i=1}^N (x_i - \bar{x})^2}$$

where

- N is the number of spatial units
- x_i is the value of a unit at location i

- $\bar{X}$ is the average of the unit values
- w_{ij} are elements of the spatial weights matrix.

Moran's I can be implemented with the 'moran.test()' function from the spdep package in R[5].

2.6.2 Tests of Spatial Dependence - Geary's C

The Geary's C statistic considers the sum of squared differences between adjacent observations or values. Values that are significantly less than 1 are an indication of rising positive spatial autocorrelation and values much higher than 1 represent increasing negative spatial correlation. Under the null hypothesis of no spatial autocorrelation, the expected value is $E[C] = 1$. The applicable formula of the statistic is

$$C = \frac{(N-1) \sum_{i=1}^N \sum_{j=1}^N w_{ij} (x_i - x_j)^2}{2 \left(\sum_{i=1}^N \sum_{j=1}^N w_{ij} \right) \sum_{i=1}^N (x_i - \bar{x})^2}$$

where

- N is the number of spatial units
- x_i and x_j are unit values at locations i and j
- $\bar{x}$ is the average of the unit values
- w_{ij} are the elements of the spatial weights matrix.

Geary's C can be implemented with the 'geary.test()' function from the spdep package in R[5].

2.6.3 Spatial Weights Matrix

The spatial weights matrix $\mathbf{W}$ is a component of the SAR model from Section 2.5. It is composed elements w_{ij} which indicate the strength or presence of a spatial connection between observations i and j . Therefore the spatial dependence term of the SAR model is represented as

$$\rho \mathbf{W} \mathbf{y} = \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j$$

where

- ρ is the spatial autocorrelation parameter; an indicator of how strongly neighboring values affect the focal observation
- $\mathbf{W}$ is the spatial weights matrix between subjects i and j with matrix elements w_{ij} that are either indicator-based ($\{0, 1\}$) or distance-based ($\frac{1}{d_{ij}}$) where d_{ij} is the distance between subjects i and j .

The indicator-based weights are

$$w_{ij} = \begin{cases} 1, & \text{if points } i \text{ and } j \text{ are neighbors} \\ 0, & \text{otherwise} \end{cases}$$

and are normalized via

$$W_{ij} = \frac{w_{ij}}{\sum_j w_{ij}}.$$

The distance-based weights are

$$w_{ij} = \begin{cases} \frac{1}{d_{ij}}, & \text{if } 0 < d_{ij} < d_{\max} \\ 0, & \text{otherwise} \end{cases}$$

and are normalized via

$$W_{ij} = \frac{\frac{1}{d_{ij}}}{\sum_j \frac{1}{d_{ij}}}.$$

- $\mathcal{N}(i)$ is the set of all neighbors of observation i . In this study, this set consists of all subjects that are a member of a cluster that is a rook neighbor of the cluster of subject i .

2.7 Data Generation

combine these sections

Given that the form of the spatial lag model contains the response on both sides of the expression, a separate closed-form expression for data generation must be considered[7][1]. This data generation expression is

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon})$$

where $\mathbf{X}\boldsymbol{\beta} = \beta x_i + \psi z_i$. x_i is an indicator of whether subject i receives the intervention and z_i is an indicator of whether subject i receives any spillover effect.

2.8 Simulation Setup

We carry out a simulation study to explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence or absence of spatial dependence, varying restrictions on clusters where spillover is permitted to occur, and the method of sampling for treatment assignment. The parameters of each simulation case are presented in Table 1. Each simulation setting is replicated for each of the spatial grid dimensions discussed previously (2x4, 3x3, and 3x4).

2.9 Simulation Algorithm

WRITE OUT THE ALGORITHM FOR SIMULATIONS:

- (1) construct spatial object
- (2) identify treatment combinations and sample data
- (3) assign intervention, control, neighbors, spillover
- (4) data generating mechanism
- (5) fit SAR model and compute parameter estimates via MLE
- (6) extract parameter estimates and aggregate
- (7) iterate for each intervention/control combination (n=10 or 50) and across all combinations (70, 126, 924)

break out the
simulation design &
results into a new
section

Figure 14 presents an example of the 8 cluster 2x4 spatial grid object. 160 subjects (20 subjects for each cluster) with uniformly distributed coordinates are sampled over the area of the spatial object and colored according to their cluster membership. Subject counts are not necessarily identical between any pair of clusters as the sample of subjects is considered over the whole space. Points in any cluster on the grid are subject to the selected treatment (intervention/control) and equally experience any potential spillover along with all other points that are members of that same cluster. Similar illustrative figures of the sampled observations over the spatial grid could be constructed and displayed for the other spatial geometries under consideration in the simulation study.

“Scenario X”

don't use sampling design
to stratify scenarios

Index	Alpha	Beta	Spillover Effect (ψ)	Spatial Dependence (ρ)	Sampling Design	Spillover Type	“scenarios”
Set 1	0.2	1.0	0.5	parameters 0.01	SRS	Control Only	
Set 2	0.2	1.0	0.6	0.01	SRS	Control Only	
Set 3	0.2	1.0	0.7	0.01	SRS	Control Only	
Set 4	0.2	1.0	0.8	0.01	SRS	Control Only	
Set 5	0.2	1.0	0.5	0.00	SRS	Control Only	
Set 6	0.2	1.0	0.6	0.00	SRS	Control Only	
Set 7	0.2	1.0	0.7	0.00	SRS	Control Only	
Set 8	0.2	1.0	0.8	0.00	SRS	Control Only	
Set 9	0.2	1.0	0.5	0.01	BSS	Control Only	
Set 10	0.2	1.0	0.6	0.01	BSS	Control Only	
Set 11	0.2	1.0	0.7	0.01	BSS	Control Only	
Set 12	0.2	1.0	0.8	0.01	BSS	Control Only	
Set 13	0.2	1.0	0.5	0.00	BSS	Control Only	
Set 14	0.2	1.0	0.6	0.00	BSS	Control Only	
Set 15	0.2	1.0	0.7	0.00	BSS	Control Only	
Set 16	0.2	1.0	0.8	0.00	BSS	Control Only	
Set 17	0.2	1.0	0.5	0.01	SRS	Control & Intervention	
Set 18	0.2	1.0	0.6	0.01	SRS	Control & Intervention	
Set 19	0.2	1.0	0.7	0.01	SRS	Control & Intervention	
Set 20	0.2	1.0	0.8	0.01	SRS	Control & Intervention	
Set 21	0.2	1.0	0.5	0.00	SRS	Control & Intervention	
Set 22	0.2	1.0	0.6	0.00	SRS	Control & Intervention	
Set 23	0.2	1.0	0.7	0.00	SRS	Control & Intervention	
Set 24	0.2	1.0	0.8	0.00	SRS	Control & Intervention	
Set 25	0.2	1.0	0.5	0.01	BSS	Control & Intervention	
Set 26	0.2	1.0	0.6	0.01	BSS	Control & Intervention	
Set 27	0.2	1.0	0.7	0.01	BSS	Control & Intervention	
Set 28	0.2	1.0	0.8	0.01	BSS	Control & Intervention	
Set 29	0.2	1.0	0.5	0.00	BSS	Control & Intervention	
Set 30	0.2	1.0	0.6	0.00	BSS	Control & Intervention	
Set 31	0.2	1.0	0.7	0.00	BSS	Control & Intervention	
Set 32	0.2	1.0	0.8	0.00	BSS	Control & Intervention	

Table 1: Simulation Scenarios for Intervention Estimation in 2x4, 3x3, and 3x4 Spatial Grid Setting

2.10 Estimation Metrics

Estimates for the baseline effect (α), intervention effect (β), spillover effect (ψ), and spatial correlation parameter (ρ) were collected for each iteration of the simulation procedure for every treatment assignment combination. Considering the initial fixed simulation parameters, the following metrics are computed for estimation of the intervention effect for the evaluation of each set of simulation conditions.

- **Bias:** a measure of how different the estimated parameter value is from the true parameter value. Small bias means that the parameter estimate is quite close to the true value.
- **Variance:** a measure of the spread of a set of estimates from multiple simulation iterations. Smaller variance indicates that the estimates are tightly grouped together and large variance means that

Algorithm 1 [EX: PLACEHOLDER] Simulation Algorithm for Intervention Effect (β) Estimation

```

1: procedure SIMULATE_SPATIAL_RCTs( $\mathcal{S}, M$ )  $\triangleright \mathcal{S}$ : Set of all 32 simulation scenarios,  $M$ : Number
   of iterations per scenario
2:   Construct Spatial Objects:
3:   Construct a spatial grid of  $N = 12$  clusters.
4:   Define the spatial weights matrix  $W$  based on cluster contiguity (e.g., Queen).
5:   Define true parameters:  $\beta = 1.0$ , and the ranges for  $\psi$  and  $\rho$ .
6:   Initialize a data structure to store final MSE and SD results for each scenario.
7:   for each scenario  $s$  in the set  $\mathcal{S}$  do  $\triangleright$  Outer loop iterates through all 32 combinations
8:     Let  $s = (\text{Sampling Design}, \rho, \psi, \text{Spillover Type})$ .
9:     Initialize an empty list  $\mathcal{L}_{\hat{\beta}}$  to store estimates from this scenario.
10:    for iteration  $m = 1, \dots, M$  do  $\triangleright$  Inner loop for Monte Carlo replicates
11:      Assign Treatment and Neighbors:
12:      Assign  $n_I = 6$  intervention and  $n_C = 6$  control clusters based on the Sampling Design.
13:      Identify the set of neighbors for each cluster using the weights matrix  $W$ .
14:      Define spillover influence according to the Spillover Type.
15:      Generate Data:
16:      Generate outcome data  $y$  for all clusters from a data generating process that includes:
        Spatial dependence (using parameter  $\rho$ ),
        Intervention effect (using parameter  $\beta$ ),
        Spillover effects (using parameter  $\psi$ ).
        Example SAR model:  $y = (I - \rho W)^{-1}(X\beta + Z\psi + \epsilon)$ .
17:      Fit Model and Estimate Parameters:
18:      Fit a Spatial Autoregressive (SAR) model to the generated data.
19:      Estimate the intervention effect parameter  $\hat{\beta}^{(m)}$  using Maximum Likelihood Estimation.
20:      Store the estimate:  $\mathcal{L}_{\hat{\beta}} \leftarrow \mathcal{L}_{\hat{\beta}} \cup \{\hat{\beta}^{(m)}\}$ .
21:    end for  $\triangleright$  End of iteration loop
22:    Aggregate Estimates for Scenario:
23:    Aggregate the stored estimates in  $\mathcal{L}_{\hat{\beta}}$  across all  $M$  iterations.
24:    Compute the Mean Squared Error (MSE) and Standard Deviation (SD) for the estimated  $\hat{\beta}$ .
25:    Record these aggregated metrics for scenario  $s$ .
26:  end for  $\triangleright$  End of scenario loop
27:  return Recorded MSE and SD for all scenarios.
28: end procedure

```

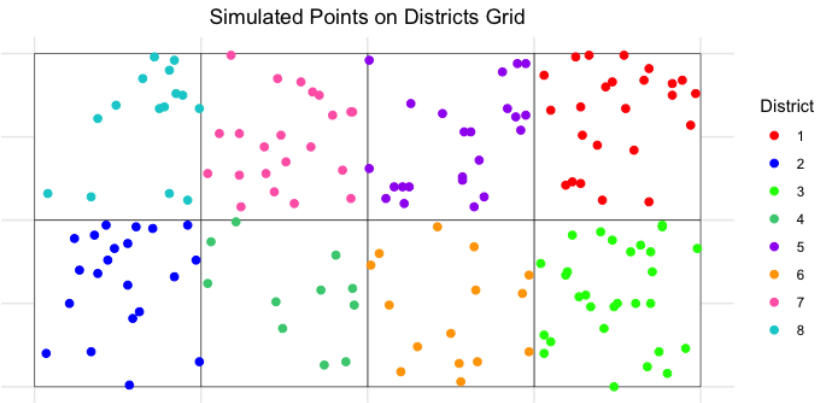
the estimates are spread out

- **Mean Squared Error (MSE):** measures the average squared difference between each estimate and the true value of a parameter. Mean Squared Error combines variance to understand the consistency of a set of estimates as well as bias to understand any trend in deviation from the true value into a measure of estimation accuracy.

Mean Squared Error is the primary metric of consideration to understand the efficient estimation of the intervention effect under all simulation settings.

3 Simulation Results

A simulation study was carried out to consider the contrast of block stratified and random treatment assignment sampling methods with consideration for spillover effects, spatial dependence, and restrictions on spillover application between intervention and control regions. The results of this study are detailed in the remainder of this section.



instead of table 1, maybe just list “in words” the parameters/ conditions

Figure 14: Example of simulated observation coordinates (n=160) on a 2x4 grid of clusters

Table 2/3/4 can speak for table 1 (not needed)

Sampling Design (2x4)	Spatial Dependence (ρ)	Spillover Effect (ψ)	Control Only Spillover		Intervention & Control Spillover	
			MSE	SD	MSE	SD
SRS	Yes (0.01)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
	No (0.00)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
BSS	Yes (0.01)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
	No (0.00)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X

consider flipping axis of sampling design & TrtSpill

Include columns for Bias for each SRS & BSS

Table 2: SRS vs. BSS: Intervention Effect Error Estimates - 2x4 Spatial Grid Setting

“Scenario”

3.1 Intervention Effect Estimates (Control Only Spillover)

3.2 Intervention Effect Estimates (Intervention & Control Spillover)

3.3 Comparison of Methods

Sampling Design (3x3)	Spatial Dependence (ρ)	Spillover Effect (ψ)	Control Only Spillover		Intervention & Control Spillover	
			MSE	SD	MSE	SD
SRS	Yes (0.01)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
	No (0.00)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
BSS	Yes (0.01)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
	No (0.00)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X

Table 3: SRS vs. BSS: Intervention Effect Error Estimates - 3x3 Spatial Grid Setting

Sampling Design (3x4)	Spatial Dependence (ρ)	Spillover Effect (ψ)	Control Only Spillover		Intervention & Control Spillover	
			MSE	SD	MSE	SD
SRS	Yes (0.01)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
	No (0.00)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
BSS	Yes (0.01)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X
	No (0.00)	0.5	X	X	X	X
		0.6	X	X	X	X
		0.7	X	X	X	X
		0.8	X	X	X	X

Table 4: SRS vs. BSS: Intervention Effect Error Estimates - 3x4 Spatial Grid Setting

figures will need to be condensed

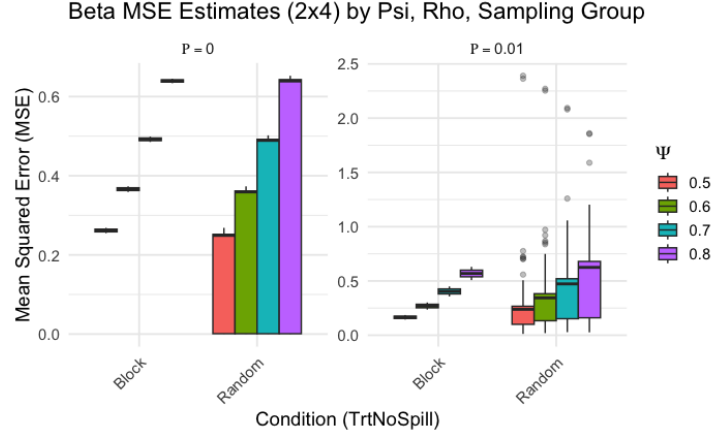


Figure 15: Intervention effect MSE for 2x4 cluster grid with spillover allowed only on control clusters

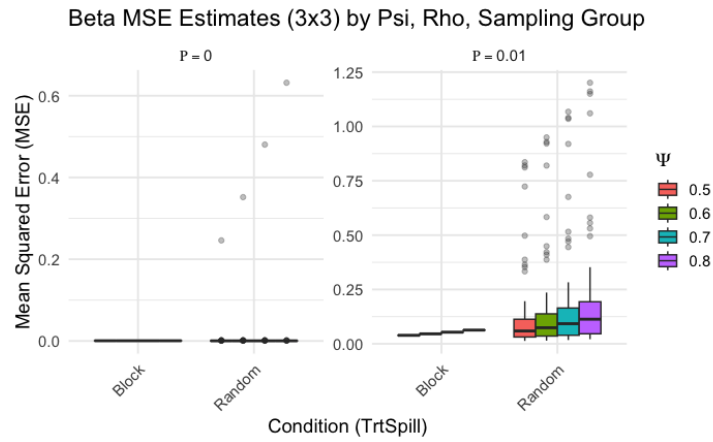


Figure 16: Intervention effect MSE for 3x3 cluster grid with spillover allowed only on control clusters

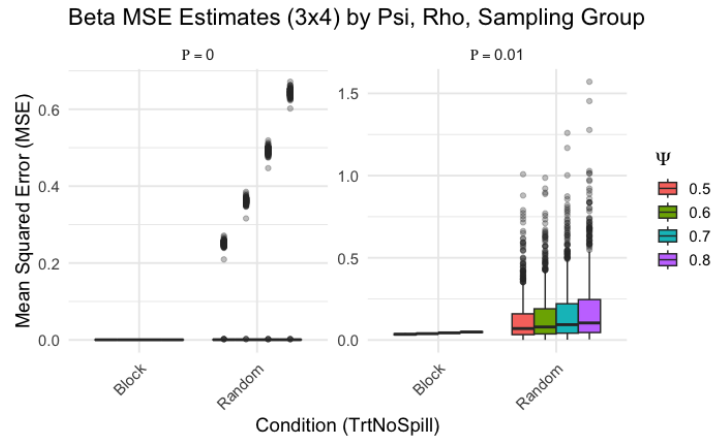


Figure 17: Intervention effect MSE for 3x4 cluster grid with spillover allowed only on control clusters

4 Application

We apply these findings to a pair of studies in development. One study is supported by staff at the Georgia Department of Community Supervision and the other is a study in development for the North Carolina Department of Adult Corrections. The investigators leading each study both seek recommendations for treatment assignment methodology.

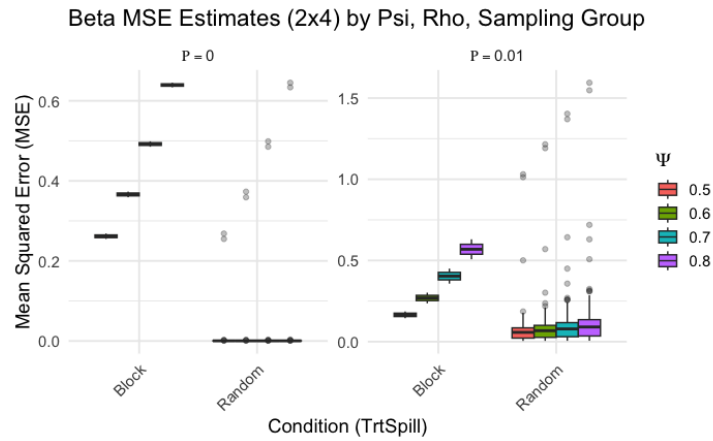


Figure 18: Intervention effect MSE for 2x4 cluster grid with spillover allowed on control and intervention clusters

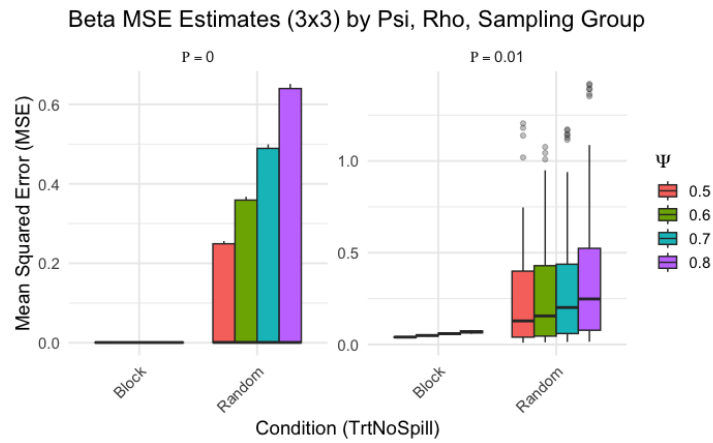


Figure 19: Intervention effect MSE for 3x3 cluster grid with spillover allowed on control and intervention clusters

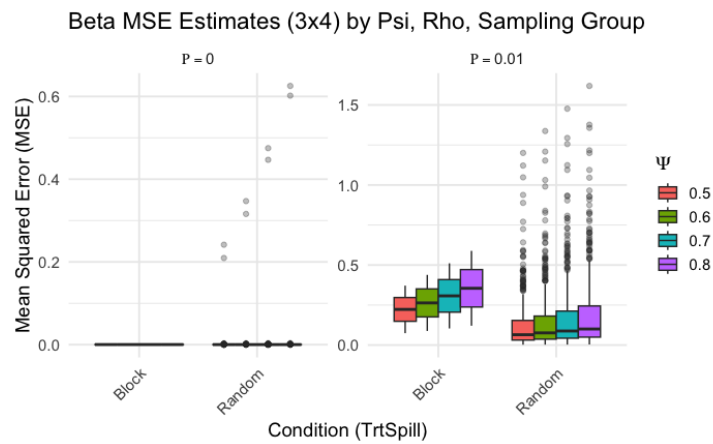


Figure 20: Intervention effect MSE for 3x4 cluster grid with spillover allowed on control and intervention clusters

4.1 GA Department of Community Supervision

4.1.1 Motivating Example

4.1.2 Recommendation BSS is robust, not optimal

4.2 NC Department of Adult Corrections

4.2.1 Motivating Example

4.2.2 Recommendation

Big emphasis on “it depends”
 - need to understand your data
 - need to define what is “important”
 (small absolute error? small variation in error)

5 Discussion

5.1 Relevance of results

- Valuable comparison between random and block stratification sampling in a variety of cases
 - spillover parameter
 - spatial dependence
 - restrictions on spillover application (treatment, control, or both)
- Understanding of effect of spatial dependence on intervention effect estimation error
- Clear relationship seen between magnitude of spillover and estimation error on intervention effect
- Other takeaways

5.2 Limitations

- Multicollinearity between direct (intervention) & indirect (spillover) effects
- Expanded simulation space becomes quickly computationally infeasible
- rudimentary spillover mechanism (binary effect) is applied. No distance-based effect decay is modeled in this simulation study.

5.3 Future Considerations

- Other neighbor structures (rook vs. other)
- Additional spillover structures (categorical, linear, exponential decay)
- Consideration of “additive” spillover (being neighbors with multiple intervention cells allows for more spillover)
- “prior” weighting of intervention based on cluster subject density
- Extension of simulation parameters

6 Appendix

[6] [9] [10] [11] [15] [16]

References

- [1] Luc Anselin. Spatial Econometrics. In Badi H. Baltagi, editor, *A Companion to Theoretical Econometrics*, pages 310–330. Wiley, 1 edition, January 2003.
- [2] Jade Benjamin-Chung, Jaynal Abedin, David Berger, Ashley Clark, Veronica Jimenez, Eugene Konagaya, Diana Tran, Benjamin F Arnold, Alan E Hubbard, Stephen P Luby, Edward Miguel, and John M Colford. Spillover effects on health outcomes in low- and middle-income countries: a systematic review. *International Journal of Epidemiology*, 46(4):1251–1276, August 2017.
- [3] Jade Benjamin-Chung, Haodong Li, Anna Nguyen, Gabriella Barratt Heitmann, Adam Bennett, Henry Ntuku, Lisa M. Prach, Munyaradzi Tambo, Lindsey Wu, Chris Drakeley, Roly Gosling, Davis Mumbengegwi, Immo Kleinschmidt, Jennifer L. Smith, Alan Hubbard, Mark Van Der Laan, and Michelle S. Hsiang. Extension of efficacy range for targeted malaria-elimination interventions due to spillover effects. *Nature Medicine*, 30(10):2813–2820, October 2024.
- [4] Roger Bivand. R Packages for Analyzing Spatial Data: A Comparative Case Study with Areal Data. *Geographical Analysis*, 54(3):488–518, July 2022.
- [5] Roger S. Bivand and David W. S. Wong. Comparing implementations of global and local indicators of spatial association. *TEST*, 27(3):716–748, September 2018.
- [6] Amy M Crisp, M Elizabeth Halloran, Ira M Longini, Gonzalo Vazquez-Prokopec, and Natalie E Dean. Covariate-constrained randomization with cluster selection and substitution. *Clinical Trials*, 20(3):284–292, June 2023.
- [7] A. Stewart Fotheringham and Peter Rogerson. *The Sage handbook of spatial analysis*. The Sage handbook of. Sage, London, 2009.
- [8] R. C. Geary. The Contiguity Ratio and Statistical Mapping. *The Incorporated Statistician*, 5(3):115, November 1954.
- [9] Christopher Jarvis, Gian Luca Di Tanna, Daniel Lewis, Neal Alexander, and W. John Edmunds. Spatial analysis of cluster randomised trials: a systematic review of analysis methods. *Emerging Themes in Epidemiology*, 14(1):12, September 2017.
- [10] Christopher I. Jarvis, Lea Multerer, Daniel Lewis, Fred Binka, W. John Edmunds, Neal Alexander, and Thomas A. Smith. Spatial Effects of Permethrin-Impregnated Bed Nets on Child Mortality: 26 Years on, a Spatial Reanalysis of a Cluster Randomized Trial. *The American Journal of Tropical Medicine and Hygiene*, 101(6):1434–1441, December 2019.
- [11] Pei-Sheng Lin, Yih-Jeng Yu, and Jun Zhu. Disease mapping for spatially semi-continuous data by estimating equations with application to dengue control. *Statistics in Medicine*, 42(20):3636–3648, September 2023.
- [12] Paula Moraga. *Spatial statistics for data science: theory and practice with R*. Chapman and Hall/CRC Data Science Series. CRC Press, Boca Raton London New York, first edition edition, 2024.
- [13] P. A. P. Moran. The Interpretation of Statistical Maps. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 10(2):243–251, July 1948.
- [14] P. A. P. Moran. Notes on Continuous Stochastic Phenomena. *Biometrika*, 37(1/2):17, June 1950.
- [15] Syarifah Diana Permai, Ronald Jauri, and Andry Chowanda. Spatial autoregressive (SAR) model for average expenditure of Papua Province. *Procedia Computer Science*, 157:537–542, 2019.

- [16] Ian W. Tang, Scott M. Bartell, and Verónica M. Vieira. Unmatched spatially stratified controls: A simulation study examining efficiency and precision using spatially-diverse controls and generalized additive models. *Spatial and Spatio-temporal Epidemiology*, 45:100584, June 2023.
- [17] Aman Ullah and David E. A. Giles. *Handbook of Applied Economic Statistics*. Number v.Vol. 155 in Statistics: a Series of Textbooks and Monographs. Chapman and Hall/CRC, Boca Raton, 1998.
- [18] Lance A. Waller. *Applied Spatial Statistics for Public Health Data*. Number v.368 in Wiley Series in Probability and Statistics Ser. John Wiley & Sons, Incorporated, Hoboken, 1st ed edition, 2004.